Nithin Surya S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.27
The Suguna Pip School	Class XII (CBSE)	2024	92.20%
Sri Chaitanya Techno School	Class X (CBSE)	2022	93.80%

INTERNSHIP

Robotics Intern | RobotoAI Technologies Pvt Ltd

- Built and tested simulation models using ROS2, Gazebo and URDF.
- Integrated a LiDAR sensor with a ROS2 robot in Gazebo and worked with its LaserScan data for perception and navigation.
- Assisted in debugging and testing an autonomous mobile robot, helping identify and resolve issues during hardware testing.
- Gained practical exposure to SLAM and Nav2 while working on autonomous navigation in simulation.

PROJECTS

Face Recognition Attendance System | Python, OpenCV, DeepFace, CustomTkinter

- Developed a desktop application for real-time face recogniton and automated attendace logging using a local face database.
- Built a guided face enrollment module with multi-pose image capture, dataset management, and an intuitive webcam interface.
- Designed a modular application seperating registration, recognition, and attendance logging for maintenance and future enhancements.

FFT-Based Trajectory Smoothing for a 2-Link Robotic Arm | Python, Numpy, Matplotlib

- Simulated a 2-link robotic arm tracing a figure-eight trajectory using inverse kinematics
- Applied FFT-based low-pass filtering to smooth noisy joint trajectories before control.
- Evaluated the effect of filtered trajectories on motion tracking using a PD-controlled joint dynamics model.

Autonomous Mobile Robot Simulation | ROS2, Gazebo

- Developed a simulated mobile robot with LiDAR-based SLAM and autonomous navigation using ROS2 and Gazebo.
- Integrated SLAM and Nav2 for map generation and goal-based navigation in simulation.
- Configured and tested the robot description, LiDAR, mapping, localization, and navigation stack.

Swarm Robotics | Python, ROS

- Developed leader and follower scripts for a leader-follower swarm robotics project.
- Implemented follower behaviour to maintain a position behind a randomly moving leader.

Cyclotron Demonstration Model | Electronics

- Built the electrical system of a cyclotron demonstration model, performing circuit wiring, soldering and prototype debugging

WALL-E Inspired autonomous Robot | Fusion 360

- Designed the robot's head and neck assembly in Fusion 360, adapting a existing neck mechanism to a single-actuator, 2-DOF configuration

Ghost Survivors | Unity, C#

- Developed a 2D survival game for the Milan fest, featuring progressively challenging enemy waves,
- Implemented enemy spawning, pathfinding, weapon mechanics, health scoring and upgrade systems

POSITIONS OF RESPONSIBILITY

Overall head, Robotix, IIT Hyderabad

Electronics & Embedded Core, Robotix, IIT Hyderabad

Mechatronics Core, Tinkerer's Lab, IIT Hyderabad

Core, Elektronica, IIT Hyderabad

Gamedev core, Glitch, IIT Hyderabad

Inbound Buddy, International Relations Cell, IIT Hyderabad

SKILLS

Languages: Python, C

Tools & Platforms: ROS2, Gazebo, Nav2, URDF, Unity, Fusion 360, Solid Edge, OpenCV, DeepFace, NumPy, GitHub, Ubuntu

Technologies: SLAM, Machine Learning, Computer Vision, Autonomous Navigation, Robot Simulation, LiDAR, PyTorch

RELEVANT COURSES

CSE: Data Structures & Applications, Introduction to Programming

Electrical: Digital Circuits

Maths: Calculus, Elementary Linear Algebra, Introduction to Probability, Introduction to Statistics

Computational Physics